Subject: Mathematics **CBSE CLASS 10 – CHAPTER TEST (SET – 4)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 7: Coordinate Geometry** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided wherever specified.
2. **Section A** comprises Multiple Choice / Objective Questions of 1 mark each.
3. **Section B** comprises Short Answer Type questions.
4. **Section C** comprises Long Answer & Competency Based questions.
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The perimeter of the triangle formed by the vertices $(0, 0)$, $(2, 0)$ and $(0, 2)$ is : [1]

(A) 4 units (B) 6 units (C) $6\sqrt{2}$ units (D) $(4 + 2\sqrt{2})$ units

Q2. If the endpoints of a diameter of a circle are $(-4, 2)$ and $(4, -6)$, then the diameter of the circle is: [1]

(A) $8\sqrt{2}$ units (B) 10 units (C) $4\sqrt{5}$ units (D) $10\sqrt{2}$ units

Q3. The ratio in which the x-axis divides the line segment joining the points $A(1, -3)$ and $B(4, 5)$ is: [1]

(A) 3 : 5 (B) 5 : 3 (C) 1 : 4 (D) 2 : 3

Q4. Two of the vertices of ΔPQR are $P(-1, 5)$ and $Q(5, 2)$. The coordinates of a point which divides PQ in the ratio 2 : 1 are : [1]

(A) $(3, -3)$ (B) $(5, 5)$ (C) $(3, 3)$ (D) $(5, 1)$

Q5. If the point $C(k, 4)$ divides the join of the points $A(2, 6)$ and $B(5, 1)$ in the ratio 2 : 3, then the value of k is : [1]

(A) 16 (B) $\frac{28}{5}$ (C) $\frac{16}{5}$ (D) $\frac{8}{5}$

SECTION B: Short Answer Type Questions (6 Marks)

Q6. In what ratio does the y-axis divide the line segment joining the points $P(-4, 5)$ and $Q(3, -7)$? Also find the coordinates of the point of intersection. [3]

Q7. Find the ratio in which the line segment joining $A(1, -5)$ and $B(-4, 5)$ is divided by the x-axis. [3]

SECTION C: Long Answer & Competency Based Questions (9 Marks)

Q8. If the points $A(6,1)$, $B(p,2)$, $C(9,4)$ and $D(7,q)$ are the vertices of a parallelogram ABCD, then find the values of p and q. Hence, check whether ABCD is a rectangle or not. **[5]**

Q9. Competency Case Study: On an annual athletics ground, markers are established at coordinates $A(1,2), B(5,5), C(9,8)$ and $D(5,5)$. **[4]**

(i) Find the distance between marker A and marker B.

(ii) Show that markers A, B and C lie in a single straight running lane.

(iii) Find the coordinates of the refreshment stall placed exactly midway along AC.

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